

Introduction to Python and Python Basics

What can Python do?

Why Python?

Python Syntax compared to other programming languages

Python Install

The print statement and Comments

Python Data Structures & Data Types

Simple Input & Output, Output Formatting

Operators in Python

Ways to String Creations

Accessing and Compare

Common methods

Python Program Flow

Indentation The If statement and its' related statement statement The while and for loop The r

Data Structures and Collections

Dictionary Comprehensions

Filter

list comprehensions

Collections

UserDict , UserList and UserString

Python

Create your own functions

Functions, Parameters, and Variable Arguments

Function Documentations

OOPS

Decorators and Generators

Creating Classes and Objects

Constructors

Inheritance

Polymorphism

Generators and Iterators

Instance Methods
Create a Module and Standard Modules
Dynamic Binding and typing
Abstraction
Iterators
Generators
The Functions any and all
With Statement
Data Compression
Decorators

File Handling

File Handling Modes
Reading Files
Writing Or Appending to Files
Handling File Exceptions

Exceptions Handling

Errors and Exceptions
Types of Exceptions
Exception handling with try
Handling Multiple Exceptions
Exception Classes
Custom Exceptions

Regular Expression and SQL

Split
Working with special characters, dates, and emails
Quantifiers
Match and find all
Character sequence and substitute
Search method

Review

Evaluation

Evaluation

Fast API

Introduction
First Steps
Path Parameters, Query Parameters, Request Body

Query Parameters and String Validations and Path Parameters
Numeric Validations
Query Parameter Models

Review

Fast API

Discussion

Static CRUD Application

Body - Multiple Parameters, Fields and Nested Models

Header Parameters and Models

Response Model - Return Type

Response Status Code

Form Data and Models

SQL (Relational) Database APIs

All CRUD APIs using FASTAPI

Review

Fast API(Gen AI)

Discussion

Integrating Basic LLM Model with FASTAPI

Implementing Basic Chatbot Application(Backend)Fast API

CRUD Application

Review